

# TeleClinic Platform — Month 3 Learning Review

**Candidate:** Twahirwa, Herve

**Role:** Data Analyst & MEL Associate · **Period:** Feb–Apr 2026 (Month 3)

**Code repository:** <https://github.com/twahih/lrembo-mel-assessment>

## Part 1: Data Quality Audit

I started with the data dictionary, then ran missingness checks, duplicate primary-key tests, and join checks across all seven tables before calculating any performance numbers. Below is every material data quality issue I found, with what I think caused it and what it means for reporting.

| # | Issue                                                    | Likely explanation                                                                                     | Impact on reporting                                                       |
|---|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 1 | 18 referrals (17.6%) with missing authorisation status   | Referral initiated clinically but authorisation never recorded or did not sync (NULL = missing record) | RM-01/02 metrics unreliable; processing time cannot be calculated         |
| 2 | 122 completed consults without notes; 169 without ICD    | Clinicians closing under time pressure; no hard stop before sign-off                                   | Clinical audit weakened; quality metrics must use completed consults only |
| 3 | 307 rows with null diagnosis/notes/ICD/call_type (27.6%) | Expected for cancelled/no-show bookings — not a data entry bug                                         | Misleading if included in clinical quality denominators                   |
| 4 | 213 incomplete registrations / failed validation         | Insurance API: NID not found, expired cover, timeouts                                                  | Onboarding funnel incomplete; 491 registered patients never booked        |
| 5 | 62 lab results uploaded but not viewed by clinician      | No alert when results sit unreviewed                                                                   | Patient safety risk — results may never reach patient                     |
| 6 | 23 lab tests without uploaded results                    | Lab capacity, sample issues, or sync delay                                                             | Lab completion and TAT metrics incomplete                                 |
| 7 | 81 prescriptions not dispensed                           | Non-collection, stock-out, or care pathway change                                                      | Care continuity gap; hard to separate clinical vs operational reasons     |
| 8 | 15 referrals not authorised + 18 unknown status          | Triage rejection or admin backlog at receiving facility                                                | Patient may not know referral was denied or stalled                       |
| 9 | failure_reason null on 987 insurance log rows            | Field only populated on failed attempts                                                                | Fine if filtered to failures; misleading if reported raw                  |

### Join problem

I checked every foreign key link (patient\_id, consultation\_id, original\_consult\_id). All IDs match cleanly — zero orphan records. The deliberate referral issue is different: 18 referrals have a valid consultation\_id but NULL authorised, authorisation\_datetime, and processing\_hours. I treated these as incomplete workflow records, not bad joins. For referral authorisation rate I used only records with known status (69 authorised out of 84 known = 82.1%), and reported the 18 unknowns separately rather than dropping them.

### Low-volume district: Musanze

Musanze stands out: only 6 consultations against 21 registrations (coverage ratio 0.29), the lowest of all 12 districts. Three plausible explanations: (1) later rollout in the 13-week window; (2) rural connectivity and awareness — all 21 Musanze patients are rural; (3) small-number volatility with only 21 registrations. To investigate I would need district activation dates, clinician roster hours, outreach logs, and network coverage data.

### Top patient safety concern

The 62 uploaded lab results that clinicians never viewed. In telemedicine the platform is the only bridge between lab and patient — a patient who tests and hears nothing may assume all is well.

## Part 2: Metrics & Analysis

I picked five metrics that together cover access, clinical quality, patient safety, care continuity, and the onboarding barrier — without duplicating the same story twice.

| Metric                             | Result            | Definition & exclusions                       | Why I chose it                                                        |
|------------------------------------|-------------------|-----------------------------------------------|-----------------------------------------------------------------------|
| Consultation completion rate       | 72.4% (807/1,114) | Completed ÷ all booked consults               | Shows whether demand converts to care; 14.9% no-show is lost capacity |
| Full clinical documentation rate   | 79.1% (638/807)   | Notes + ICD both Yes; completed consults only | Care that cannot be audited is hard to govern                         |
| Lab result clinician review rate   | 82.3% (288/350)   | Clinician viewed = Yes; uploaded results only | Unreviewed results are a direct safety gap                            |
| Prescription dispensing rate       | 84.5% (440/521)   | Dispensed Yes ÷ all prescriptions             | Remote prescribing only helps if pharmacy dispenses                   |
| First-attempt insurance validation | 82.2%             | Success on attempt_number = 1                 | Front-door barrier; Uninsured succeeds only 9.9%                      |

### What this data cannot tell us

Clinical outcomes, patient satisfaction, whether referred patients reached facilities, clinical appropriateness of diagnoses/Rx, or whether notes are clinically adequate.

### Result that concerns me most

Roughly one in five care episodes has incomplete documentation or an unreviewed lab result. The 62 unreviewed results are the clearest danger — telemedicine has no fallback if the clinician does not open the result. The platform is being used (1,114 bookings) but has not closed the loop on results management.

## Part 3: Equity Analysis

### Equity measures calculated

**1. Rural Representation Index (RRI):** rural share ÷ 83% national rural share. Registrations and completed consults: 60.1% rural → RRI = 0.72.

**2. District coverage ratio:** consultations ÷ registrations by district. Musanze 0.29 to Rwamagana 1.09.

Gender near-balanced (50.2% male registrants). USSD ~41% of rural consults. Insurance sharpest gap: Uninsured validation 9.9% vs 78–90% formal schemes.

Gates Foundation quarterly report paragraph

After three months, TeleClinic registered 1,200 patients and completed 807 consultations across 12 districts. Rural users account for 60% of activity vs ~83% nationally (RRI = 0.72). USSD carries ~41% of rural consults — encouraging for feature-phone users — but urban districts especially Gasabo dominate volume. Gender balanced. The equity gap I would flag: uninsured patients succeed on validation less than 10% of the time. District coverage uneven (Musanze 0.29). **Recommended action:** USSD outreach in five lowest-coverage districts plus 72-hour provisional access for Uninsured patients; track RRI monthly toward 0.85 by Month 6.

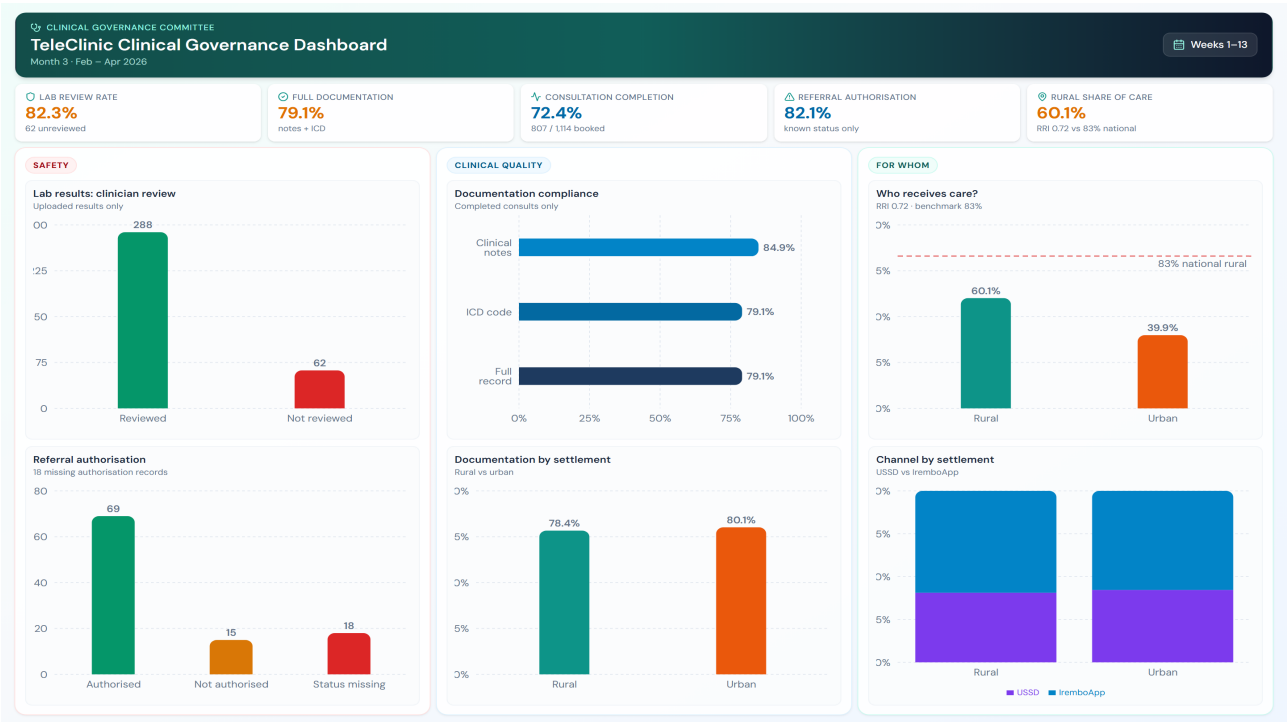
Equity dimension we cannot measure

Socioeconomic vulnerability — insurance scheme but not income or ability to pay. Add optional household quintile at registration and track validation failure → consult conversion.

Part 4: Dashboard Design

**Audience:** Clinical Governance Committee · **Central question:** Is the platform delivering safe, quality care — and for whom?

One-page React dashboard (teleclinic-dashboard/ in repo). Layout: KPI strip + Safety | Clinical Quality | For Whom. Screenshot below.



| Element                | Why this chart                                  | Decision it supports                    |
|------------------------|-------------------------------------------------|-----------------------------------------|
| KPI strip              | Five numbers give quick safe/quality/reach read | Whether Month 3 needs intervention now  |
| Lab clinician review   | 62 unreviewed shown as count not just rate      | When to escalate to clinical supervisor |
| Referral authorisation | Authorised / Not authorised / Missing (18)      | Fix workflow before scaling referrals   |

| Element                     | Why this chart                            | Decision it supports                        |
|-----------------------------|-------------------------------------------|---------------------------------------------|
| Documentation compliance    | Notes, ICD, full record side-by-side      | Require ICD before consult close?           |
| Documentation by settlement | Quality linked to rural vs urban          | Do gaps cluster geographically?             |
| Settlement share + 83% line | Rural under-representation visible        | Shift roster or outreach to rural districts |
| Channel by settlement       | USSD vs IremboApp in rural and urban care | Invest in USSD-first workflows?             |

**Metric excluded:** First-attempt insurance validation (82.2%) — onboarding metric for Gates/MOH ops, not Clinical Governance.

## Part 5: Learning Review Brief

For the Gates Foundation quarterly call

### Platform Performance at Month 3

- **People are using it.** 1,200 registrations, 1,114 bookings, 807 completed (72.4%).
- **Pathways partly close, safety gaps remain.** 84.5% Rx dispensed; 62 lab results never reviewed.
- **Rural reach below parity.** 60.1% rural vs ~83% national; Musanze lags.

### What We Cannot Yet Conclude

Whether telemedicine improves outcomes; whether referrals reach facilities; whether poorest benefit most; whether volume is sustainable — 13 weeks too short.

### One Recommended Change

Automated pathway closure tracker: flag unreviewed labs after 24h, undispensed Rx after 48h, unauthorised referrals after 72h — escalate to clinical supervisor.

### Priority Question for Months 4–6

Does closing the lab and referral loop reduce care discontinuation among rural USSD users?

## Reflection: North Star Metric

**Documented Care Completion Rate (DCCR):** share of booked consultations reaching Completed with notes and ICD both recorded. Month 3:  $638 / 1,114 = 57.3\%$ .

Completion alone (72.4%) overstates success; documentation among completed (79.1%) hides bookings that never became care. DCCR combines both. Unlike lab or dispensing rates it applies to every consult; unlike RRI it measures delivery quality not reach.

Blind spots: no check on clinical correctness, unreviewed labs (62), undispensed Rx (81), equity-blind. Still, 57.3% honestly describes Month 3: live and used, not yet reliably delivering governable care at scale.

## Process Note

I prioritised the data quality audit before performance numbers — unreviewed labs and the referral gap shaped the whole review. I spent less time on follow-ups; skipped forecasting. Most uncertain about Musanze. Built notebook + React dashboard for Part 4 governance view. With another day I'd add a pathway-closure metric chaining consult → lab → review → Rx.